Network Design for ABC Company

This comprehensive network design for ABC Company outlines a robust, scalable architecture. It addresses addressing, media selection, device choices, wireless coverage, and environmental considerations. The plan ensures efficient connectivity across three buildings while allowing for future expansion.

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Network Addressing Strategy

Building A

Subnet: 172.16.0.0/24. Supports 300 hosts with room for growth. Gateway: 172.16.0.1.

Building B

Subnet: 172.16.2.0/25. Accommodates 200 hosts with extra capacity. Gateway: 172.16.2.1.

Building C

Subnet: 172.16.3.0/25. Fits 100 hosts efficiently. Gateway: 172.16.3.1.

Future Expansion

Reserved subnets: 172.16.4.0/24 to 172.16.6.0/24. Allows for three additional buildings.



Layer 1 Media Selection

Inter-Building Connections

Fiber optic cable chosen for its high bandwidth and long-distance capabilities. Supports 1.12 km maximum distance between buildings. Offers future-proof speeds and minimal signal degradation.

Intra-Building Wiring

Cat 6 Ethernet cable selected for internal connections. Supports speeds up to 10 Gbps. Provides reliable performance for modern office networks. Easy to install and maintain.

Layer 2 Device Selection

1

Cisco 2960 Series Switches

Cost-effective solution supporting gigabit speeds. Ideal for managing local traffic in each building.

2

Key Features

Power over Ethernet (PoE) support. Advanced QoS capabilities for voice and video traffic.

3

Deployment Strategy

One switch per floor in each building. Stacking capability for easier management.



Layer 3 Device Selection

1

Intra- Gateway

Cisco 2900 Series ISR deployed in Building A. Connects to ISP for 30Mbps internet speed.

2

Inter-Building Routing

Layer 3 switches implement routing between buildings. Ensures efficient traffic flow across subnets. Wireless device WRT300N router

3

Security Features

Integrated firewall and VPN capabilities. Enables secure remote access and protects network perimeter.





Wireless Network Design

Coverage

One Cisco Aironet 1830 series AP per floor. Ensures comprehensive Wi-Fi coverage throughout buildings.

Performance

Supports 802.11ac for high-speed Wi-Fi. Delivers up to 867 Mbps theoretical data rates.

Security

WPA3 encryption and Cisco CleanAir technology. Protects against wireless threats and interference.

Each building contains various Wi-Fi client devices for example, smart-phones, tablets, and laptops



Environmental Considerations

Temperature	18-27°C (64-80°F)
Humidity	45-55% relative humidity
Cooling	CRAC units with N+1 redundancy
Power	UPS systems with 30-minute runtime
Cabling	Shielded to prevent EMI/RFI

Network Topology and Scalability

1

Core Layer

Central router in Building A. Handles inter-building routing and internet connectivity.

2

Distribution Layer

Layer 3 switches in each building. Manages traffic between access layer and core.

3

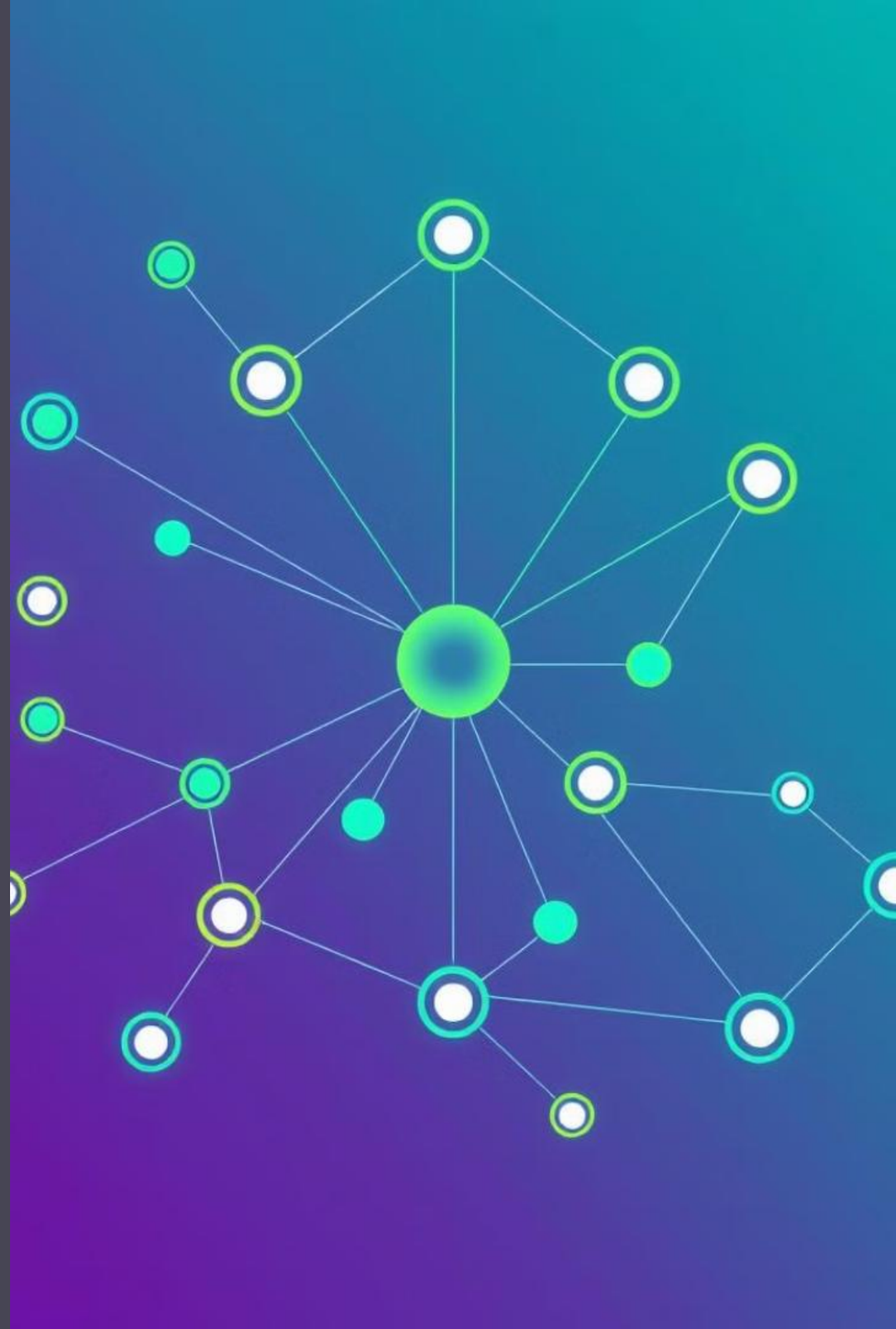
Access Layer

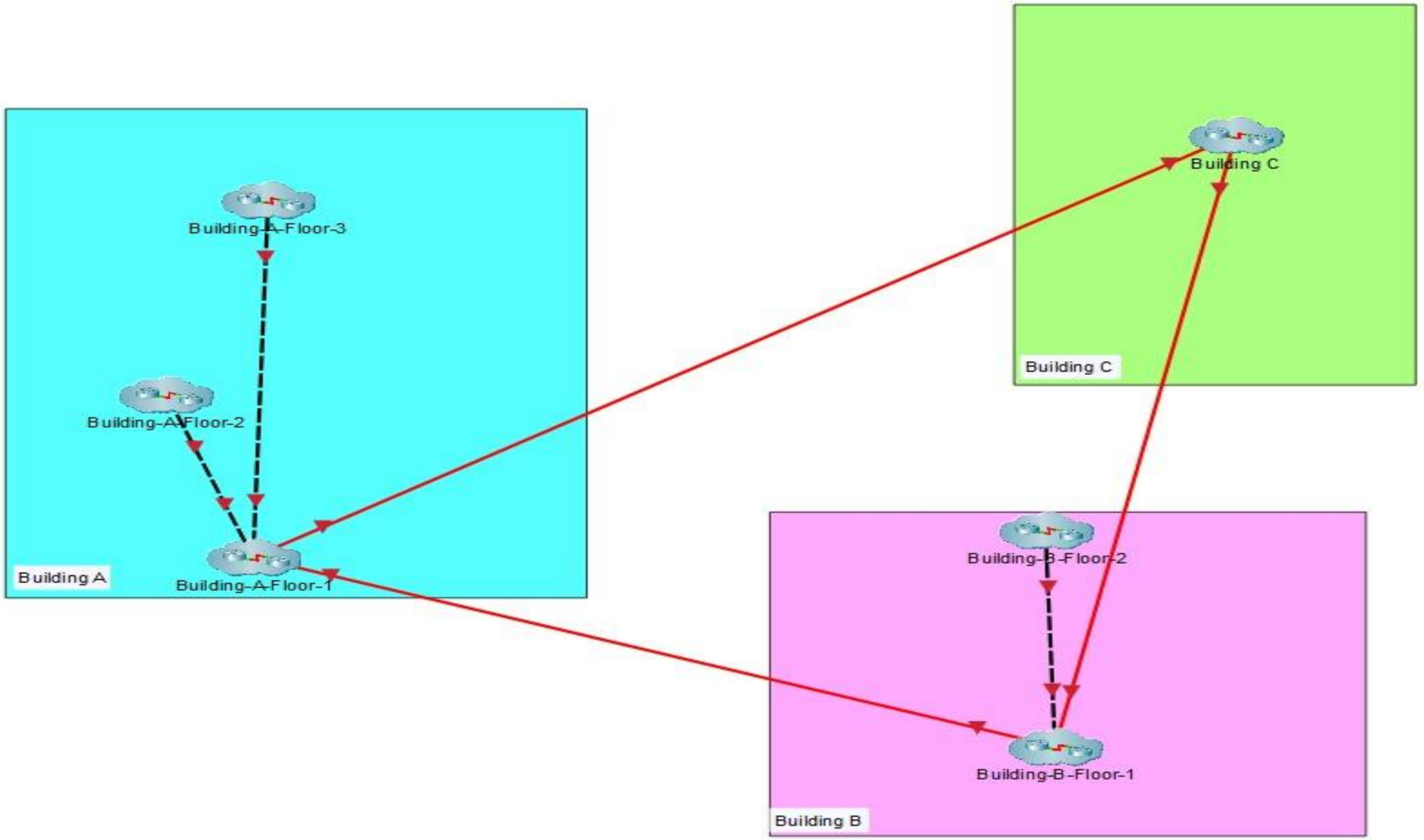
Switches and APs on each floor. Provides end-user connectivity and Wi-Fi access.

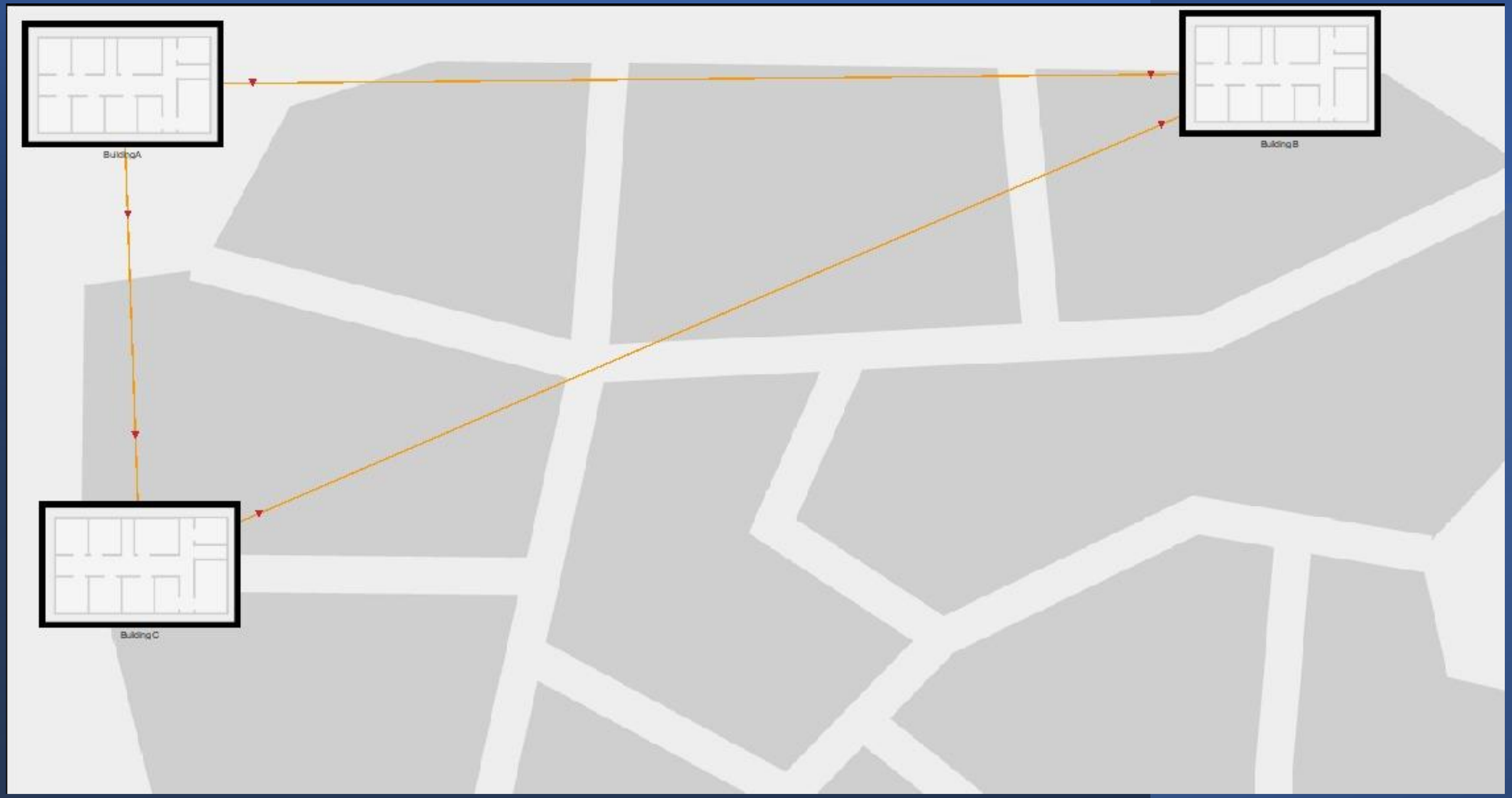
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Future Expansion

Additional subnets and core router capacity. Allows for seamless integration of new buildings.




Building C
Building C
Building-A-Floor-3
Building-A-Floor-2
Building A
Building-A-Floor-1
Building-B-Floor-2
Building-B-Floor-1
Building B


Building A
Building B
Building C


THANK YOU